Universitat Pompeu Fabra Barcelona

AES P3. Microphones: Directivity, delay alignment and differential IR

The aim of this laboratory is:

- To measure and inspect some microphone specifications such as directivity for different polar patterns.
- To estimate the differential IR, from a reference microphone and a SHURE SM57, to convolve it with test signals recorded with the reference microphone, virtualizing the SHURE SM57 response.

To complete this lab you will need:

- a KRK loudspeaker
- REW as measuring software
- a measuring microphone
- an audio interface
    - two balanced inputs for Bi-Channel Comparison
- some cables
    - 2 TRS to TRS cable as loopback-reference
    - 2 XLR cable



Fig. 1. Image of the setup.

First of all you need to prepare the Bi-channel Comparison setup.

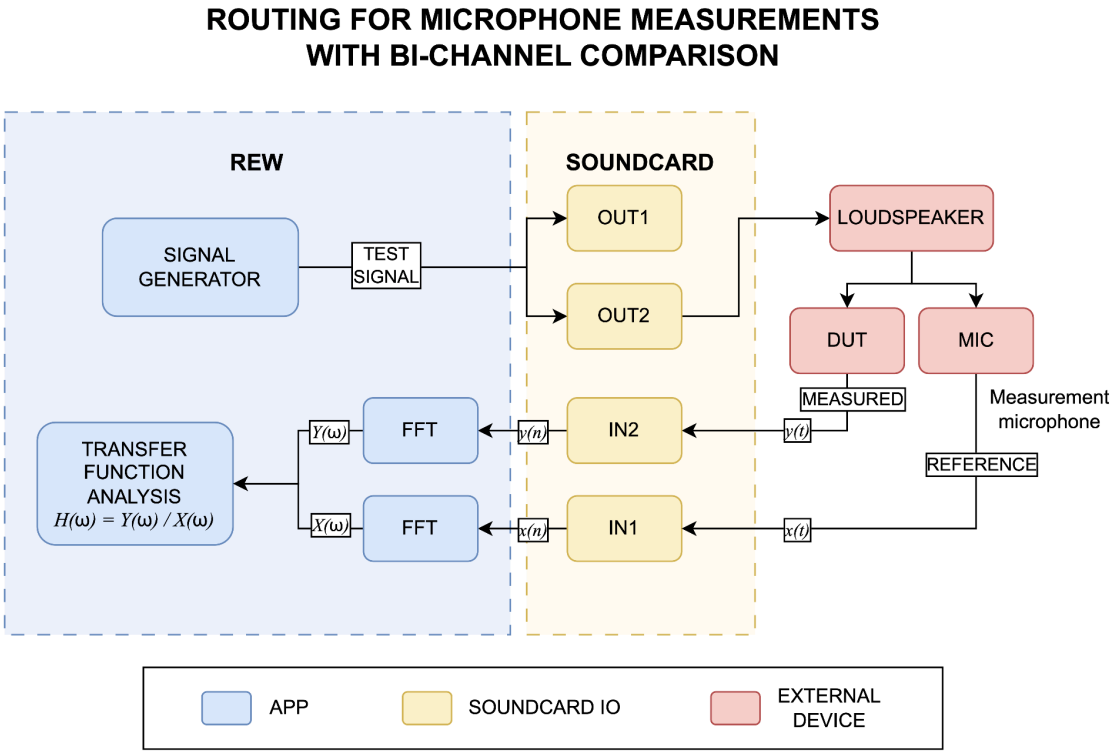


Fig. 2. Block diagram of the setup

Remember to calibrate the complete system.

1. Directivity

The goal of this section is to capture the directivity response of a microphone measured with REW, and compare it with the one provided by the manufacturer. We will use the AKG P420 in cardioid, omnidirectional and bidirectional configurations.

- Open REW software and go to the PREFERENCES menu.
- Assign the soundcard device to the Output Device and Input Device.

Use the following configurations (reference for the computer at lab 54.025) :

Drivers: Java
Output device: *Altavoces (Focusrite USB Audio)*
Output: *SPEAKER L+R*
Timing reference output: *L*

Input device: *Analogue 1+2 (Focusrite USB Audio)*
Input: *INPUT 2*
Loopback input: *INPUT 1*

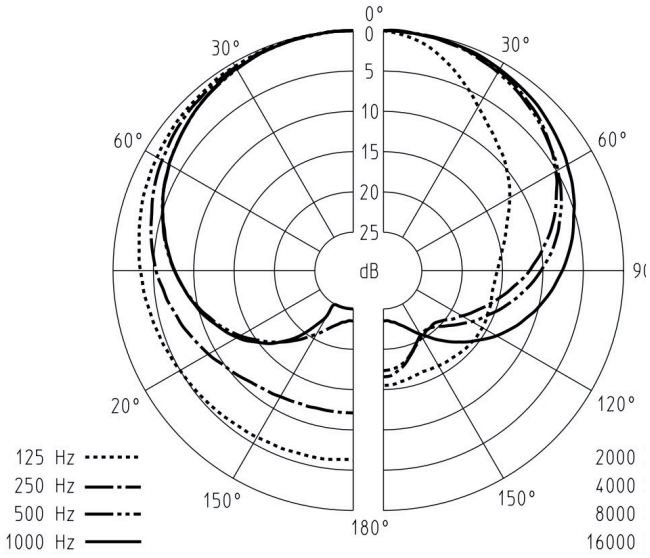


Fig. 3. Directivity patterns of Sennheiser D421

Fig. 3 shows the directivity pattern of a Sennheiser D 421. Notice how, as the diaphragm of the microphone is symmetric, the manufacturer only provides the curves from 0º to 180º for each frequency. So, we will carry out the measurements every 15º approx. from 0º to 180º rotating the microphone progressively from its base, with respect to the axis with the loudspeaker. Then, for the representation of the curves, we will mirror the response to obtain the symmetric values at 360º, obtaining a representation like the one shown in Fig.4.

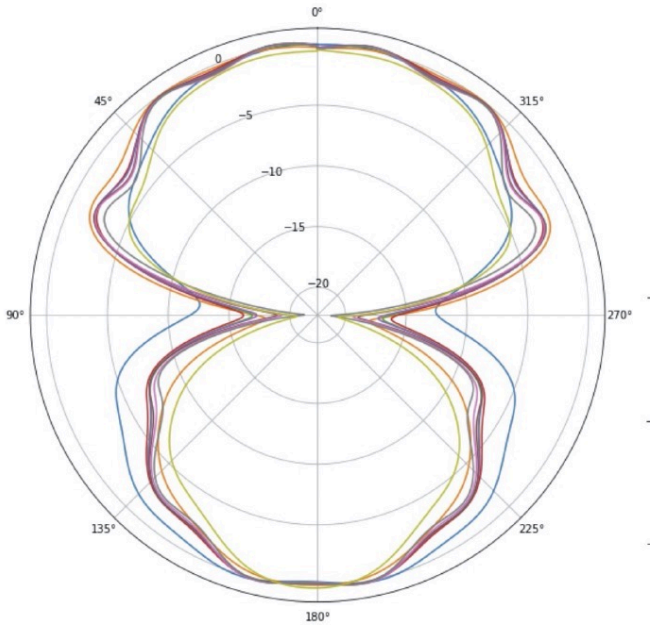


Fig. 4. Directivity patterns for Neumann TLM 107 (bi-directional).

1. Calibrate the system.
2. Place the AKG microphone 0.5 meters far from the loudspeaker. Try to reduce the influence of the early reflections that could bias the measurements using the acoustic panels.
3. Select the omnidirectional pattern for the first measure and make measurements every 15º from 0º to 180º.
4. Once all angles are measured save the measurements and export them as TXT files in *File-> Import->Export all measurements as txt*. We will use the default exporting parameters.
5. Repeat the process for each polar pattern.
6. Implement a function to display the polar patterns following the steps described in this notebook. You can start working with these measurements obtained at the Lab for the bi-directional configuration of the AKG P420.
7. Compare and comment your results with the polar patterns provided by the manufacturer.

QUESTIONS

1. Do the measurements obtained correspond to those expected (yes/no/why)?
2. Explain briefly the differences.

2. Application of the differential impulse response (differential IR) for frequency response compensation

The differential IR is the principle used in some modern audio solutions for microphone and loudspeaker virtualization, see some examples below:

- https://slatedigital.com/ml1-modeling-microphone/
- https://www.uaudio.com/microphones/sphere-l22-modeling-microphone.html?srsltid=AfmBOooaG9FnqbkpsuduSCbJHaqtbaNOpqxM6Tt11Ca5sY4lMgBaLSj2
- https://www.antarestech.com/products/vocal-effects/mic-mod
- https://www.kemper-amps.com/profiler/overview
- https://www.celestionplus.com/
- https://www.sonarworks.com/soundid-reference
- https://www.dirac.com/online-store/dirac-live-room-correction-suite/

In this section, we want to simulate the response of a specific microphone (m2) in recordings or captures done with another microphone (m1). By obtaining the differential IR of m2 (the one that we want to simulate) and m1 microphones, and convolving it with m1 microphone captures, we can “equalize” or imitate the response of m2. This could be interesting to virtualize vintage and popular microphones, or achieve specific responses. For it, we use the following operation specific to LTI systems:

$$m_{1->2}(t) = m_1(t) * h_d(t)$$

where,

- $m_{1->2}(t)$ refers to the m_1 equalized to sound as m_2, the output of the system
- $m_1(t)$ refers to the m_1, the input of the system.
- $h_d(t)$ refers to the differential IR between m_1 and m_2

According to the convolution theorem, we can obtain the magnitude and phase response by multiplication in the frequency domain,

$$M_{1->2}(\omega) = M_1(\omega) \cdot H_d(\omega)$$

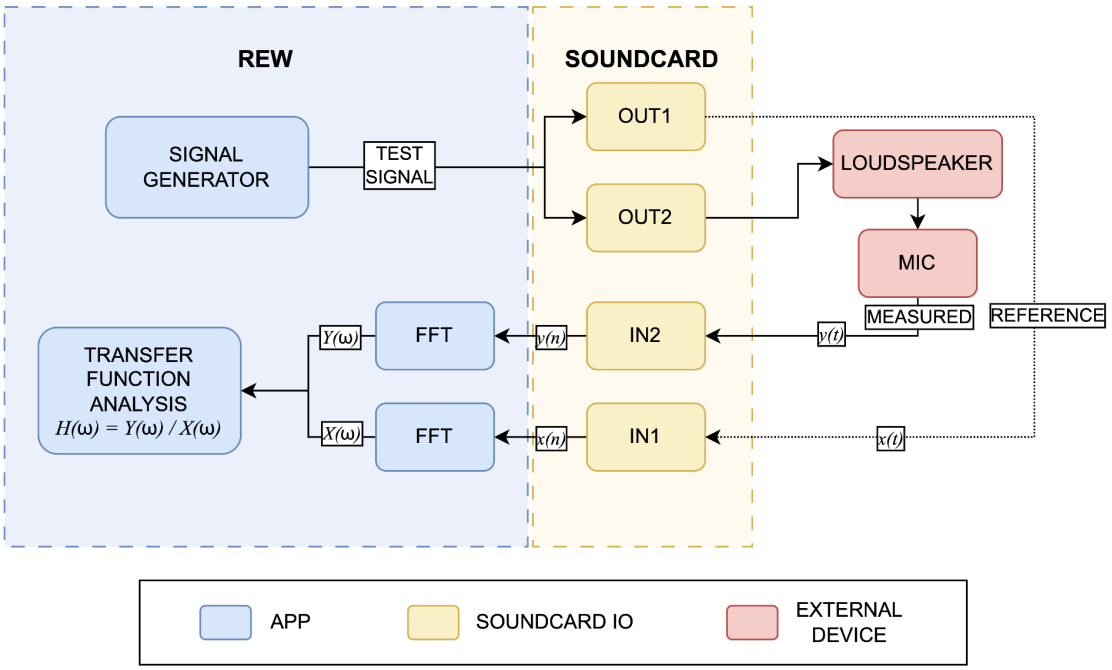


Fig. 5. Bloc diagram of the setup

1. Using the soundcard calibration with the loopback and delay compensation, make an on-axis measurement in REW for each microphone at 30cm from the loudspeaker.
    a. NOTE: APPLY DELAY COMPENSATION!
    b. PLACE MICS in the same placement!
2. Name each measurement according to the microphone (m1 for dBX and m2 for SM57).



3. To estimate $h_d(t)$ we can do it on the *ALL SPL* display. On the *Actions/Trace Arithmetic* tool, assign m_1 and m_2 to the measure B and A respectively. Check the Fig.6.
4. Select the *A/B* operation to obtain the $M_{1->2}(\omega)$ and generate.

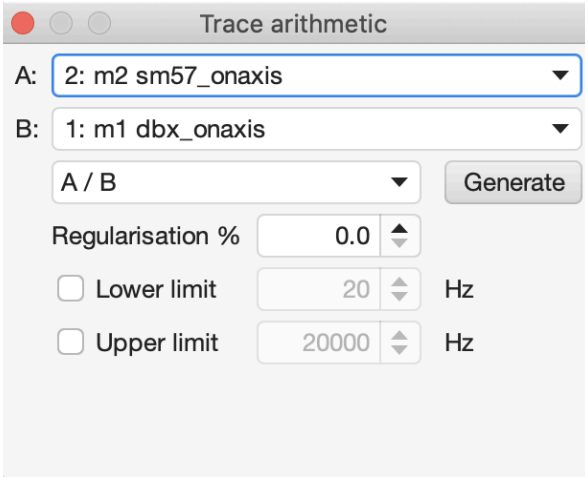


Fig. 6. Trace arithmetic to obtain the differential IR.

5. A new *A over B* measure is generated, the differential IR (sm57-to-dbx in *Fig.7*). Export it as a WAV file in File -> Export -> Export impulse response as WAV.

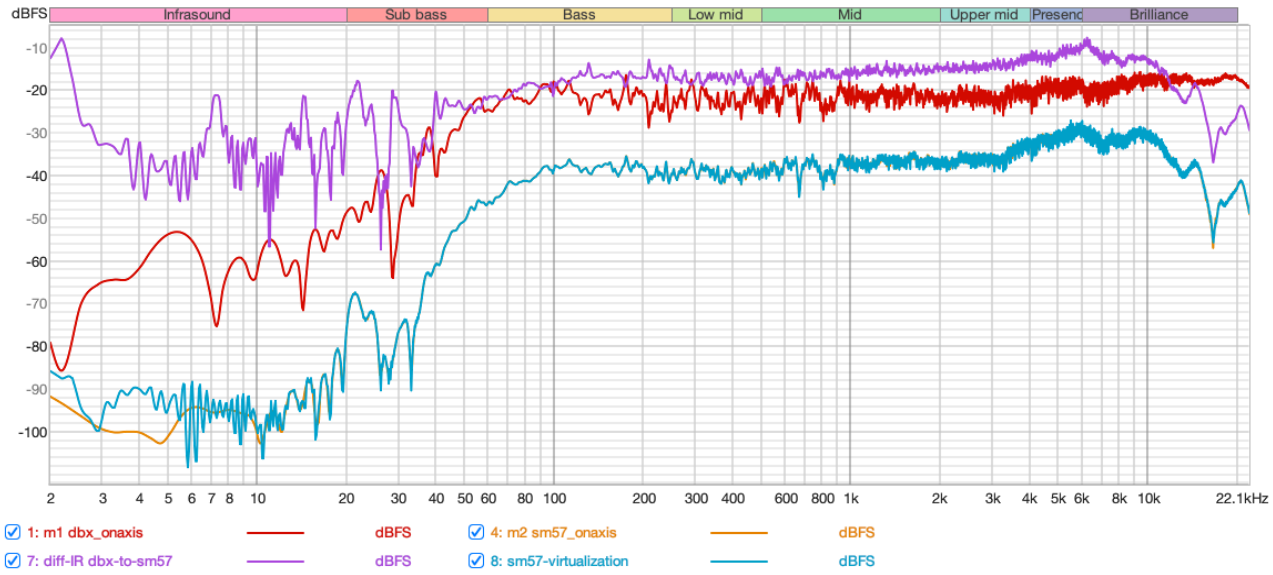


Fig 7. Magnitude frequency response for m1 and m2 microphone, the resulting differential IR and the SM57 virtualization.

6. Finally we need to capture some audio files to apply the differential IR with the m1 microphone (dBX) in the same location.
    a. Check the saved IR parameters (REW's sample rate).

b. Close REW and open REAPER.
c. In REAPER preferences, set the frequency rate to match the properties of the recordings and convolved files..
d. Create a new track and load the test file (in the link below). Reproduce the test file through the loudspeaker and simultaneously capture it with the *m1* microphone. Save the recorded file as *m2_reference.wav*

https://drive.google.com/file/d/1XFezMI2BdUGMRQfMRiKrBKloXnZBw8Xt/view?usp=sharing

QUESTIONS

1. In a Jupyter Notebook or a Python script, load the differential impulse response recorded as a WAV file and using the *numpy* package implement a function to convolve the *m2_reference.wav* with the differential IR. You should hear the captures of the reference microphone (dBX) equalized by the frequency response of the Shure SM57. Provide the results.
2. In which other areas of study do you think the differential impulse response might be useful?